Exploratory Spatial Statistical Analysis of Migratory Species Observation Patterns in India

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***Abstract*—This paper presents an exploratory spatial statistical analysis of migratory species observation patterns across India. Spatial preprocessing and exploratory spatial data analysis techniques are employed to examine spatial heterogeneity and spatial stationarity in the dataset. Local spatial autocorrelation is analyzed using Local Moran's I, revealing significant regional variation and spatially non-stationary relationships in migratory species observation density across the study area.**

***Index Terms*—Spatial statistics, migratory species, spatial heterogeneity, spatial stationarity, Moran's I, LISA**

I. Introduction

The increasing availability of large-scale geospatial datasets has enabled detailed analysis of spatial patterns across diverse application domains, including ecology, environmental monitoring, and wildlife conservation. Data related to migratory species observations are of particular interest, as migratory behavior is inherently spatial and seasonal in nature. Such datasets often exhibit strong spatial structure influenced by ecological corridors, habitat availability, seasonal movement, and observation effort.

Migratory species observation data frequently violate the assumption of spatial independence that underlies many classical statistical methods. Observations recorded in nearby locations are often spatially dependent due to shared environmental conditions or migration routes, while observation density may vary substantially across regions and seasons. These characteristics necessitate the use of spatial statistical techniques capable of explicitly accounting for spatial dependence and regional variation.

Spatial statistical analysis provides a principled framework for identifying, quantifying, and interpreting spatial patterns in georeferenced data. Two spatial properties are particularly relevant in this context: spatial heterogeneity and spatial stationarity. Spatial heterogeneity refers to the presence of non-uniform patterns across space, where statistical characteristics vary between regions. Spatial stationarity, on sub-regional the other hand, examines whether the spatial relationships governing the data remain consistent across the study area or vary locally.

Exploratory Spatial Data Analysis (ESDA) techniques are well suited for such investigations, as they allow researchers to examine spatial patterns without imposing strong model assumptions. Global measures of spatial autocorrelation summarize overall spatial dependence, while local indicators, such as Local Moran's I, identify localized clusters and spatial outliers.

In this study, we conduct an exploratory spatial statistical analysis of migratory species observation patterns across India. Using spatial preprocessing, heterogeneity analysis, and local spatial autocorrelation techniques, we examine whether migratory species observation density exhibits spatial clustering and whether the underlying spatial relationships are stationary across the study region. The results provide insights into the spatial structure of migratory observation data and inform the suitability of more advanced spatial and spatio-temporal modeling approaches.

II. Dataset Description and Pre-processing

The dataset used in this study consists of georeferenced point observations of migratory species distributed across India. Each observation represents a recorded occurrence of a migratory species and is associated with geographic coordinates (latitude and longitude) along with descriptive attributes, including observation density, taxonomic grouping, temporal information (year and month), and administrative region identifiers such as state or province.

A. Dataset Characteristics

The dataset exhibits several characteristics that make it well suited for spatial statistical analysis:

- Explicit geographic coordinates, enabling spatial referencing and neighborhood-based analysis.
- A continuous numeric variable representing migratory species observation density, suitable for spatial autocorrelation measures.
- Uneven spatial distribution across regions, reflecting migratory routes, seasonal habitats, and observation effort.
- A sufficiently large number of observations to support both global and local spatial statistical techniques.

These characteristics motivate the application of exploratory spatial data analysis (ESDA) techniques to investigate spatial dependence, heterogeneity, and stationarity in migratory species observation patterns. However, we have to note here that a lot of the values here are sparse, as shown by Figure 1, which also urges the need for aggregated features, which are denser, for better regression.

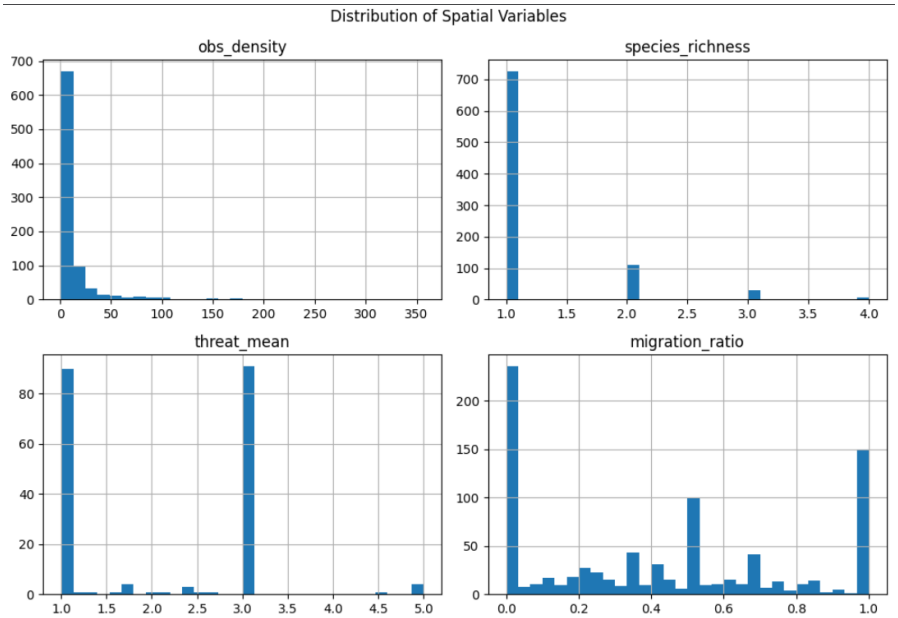

Fig. 1. Distribution of Spatial Variables in the Dataset.

B. Pre-processing Steps

Pre-processing was carried out using Python to ensure data consistency and analytical readiness prior to spatial analysis. The key steps included:

- Removal of records with missing or invalid geographic coordinates.
- Conversion of the dataset into a geospatial point format compatible with GIS software.
- Verification and standardization of attribute data types.
- Aggregation and normalization of observation-related variables where required.

Following pre-processing, the dataset was imported into QGIS and re-projected from a geographic coordinate system (EPSG:4326) to a projected coordinate system (EPSG:3857). This reprojection was necessary to enable accurate distance-based spatial analysis. The final cleaned and projected dataset served as the input for all subsequent spatial statistical analyses.

C. Aggregated Features Visualized

As part of the pre-processing workflow, the raw point data was aggregated into spatial grid cells. Three primary aggregated features were derived and mapped to visually inspect their spatial distributions: Observation Density, Species Richness, and Migration Ratio.

Figure 2 illustrates the Observation Density (*obs_density*) across the study area. The visual distribution highlights localized areas of intense observation effort, often clustered around accessible ecological zones, while large portions of the map exhibit sparse data collection.

Figure 3 displays the Species Richness (*species_richness*) per grid cell. Unlike raw observation density, species richness reveals broader ecological zones, with higher values concentrating in known biodiversity hotspots such as the coastal regions and northern foothills.

Figure 4 depicts the Migration Ratio (*migration_ratio*), representing the proportion of migratory species observations relative to total observations. This feature exhibits a more dispersed spatial pattern, capturing seasonal avian movement corridors that do not strictly align with the raw observation density.

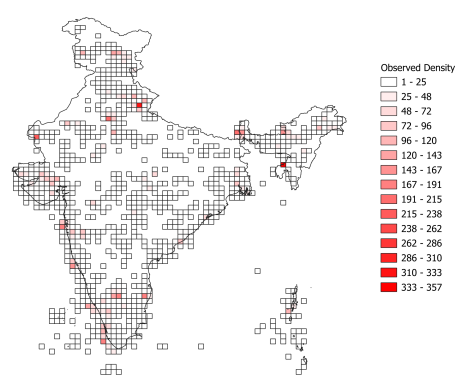

Fig. 2. Spatial distribution of aggregated Observation Density across the study grid.

III. Spatial Statistical Analysis

This section presents the spatial statistical analysis performed on the dataset to examine spatial heterogeneity and spatial stationarity in migratory species observation patterns.

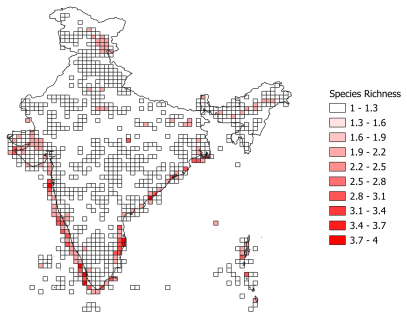

Fig. 3. Spatial distribution of aggregated Species Richness highlighting biodiversity concentration.

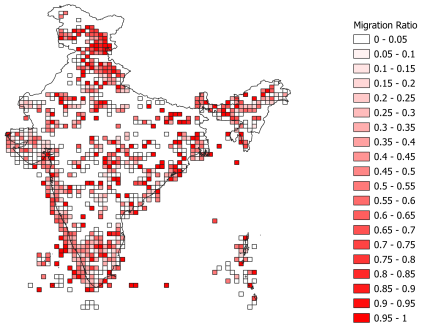

Fig. 4. Spatial distribution of aggregated Migration Ratio showing proportional migratory movement.

A. *Spatial Heterogeneity Analysis*

Spatial heterogeneity refers to the presence of non-uniform spatial patterns across a study region, where the statistical properties of a variable vary from one location to another. To visually assess heterogeneity, a graduated point map of migratory species observation density was generated (Fig. 5). The map reveals pronounced regional differences, with clusters of high observation density concentrated along specific geographic regions that may correspond to migratory corridors or seasonal habitats, while other regions exhibit sparse distributions.

To quantitatively assess spatial heterogeneity, descriptive statistics were computed by administrative regions using the "Statistics by Categories" operation in QGIS. Thus, this demonstrates substantial variation in the range, minimum, and maximum values of observation density across regions. This regional variability confirms the presence of spatial heterogeneity in migratory species observations.

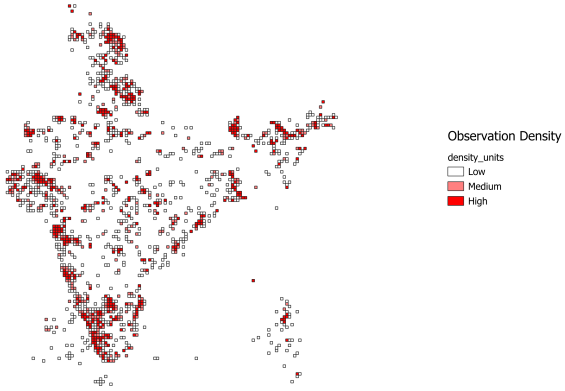

Fig. 5. Spatial heterogeneity of biodiversity observation density across India. Regions with high, medium, and low observation density indicate strong spatial variability in data distribution.

B. *Local Spatial Autocorrelation and Stationarity*

While spatial heterogeneity highlights regional variation in migratory species observation density, spatial stationarity examines whether the strength and nature of spatial relationships remain consistent across the study area. To investigate spatial stationarity, local spatial autocorrelation was analyzed using Local Moran's I through the Spatial Autocorrelation Map (LISA) approach.

To enable statistically meaningful local autocorrelation analysis, individual point observations were first aggregated into spatial polygon units representing observation density. Local Moran's I was computed on these polygonal spatial units rather than on raw point observations, allowing for well-defined neighborhood relationships and reducing the influence of point-level noise. Although these spatial units appear visually regular, they represent aggregated polygonal units rather than a uniform grid.

A k-nearest-neighbors spatial weights matrix was employed to model local spatial interactions between neighboring polygon units. Local Moran's I statistics were computed for each spatial unit, along with corresponding Z-scores and p-values to assess statistical significance.

The resulting LISA cluster map reveals that statistically significant local spatial autocorrelation is confined to specific regions, while large portions of the study area exhibit weak or insignificant local dependence. Both the magnitude and significance of the local Moran's I statistics vary spatially, indicating the presence of localized clusters as well as spatial outliers.

This spatial variation demonstrates that the relationship between migratory species observation density and spatial location is not constant across India. Consequently, the underlying spatial process governing migratory observation patterns is spatially non-stationary, highlighting the limitations of global models that assume uniform spatial dependence across the study region.

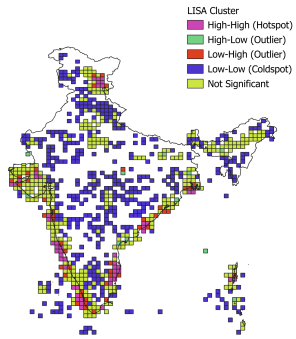

Fig. 6. LISA cluster map (Local Moran's I) of observation density across India, showing high–high (HH) and low–low (LL) spatial clusters as well as high–low (HL) and low–high (LH) spatial outliers.

TABLE I
DISTRIBUTION OF LOCAL MORAN'S I CLUSTER TYPES

Cluster Type	Number of Locations
High–High (HH)	201
Low–Low (LL)	805
High–Low (HL)	148
Low–High (LH)	409

Figure 6 presents the Local Indicators of Spatial Association (LISA) cluster map computed using Local Moran's I. The map reveals the presence of statistically significant high–high (HH) and low–low (LL) clusters, indicating localized regions of strong positive spatial autocorrelation in observation density. In addition, several high–low (HL) and low–high (LH) spatial outliers are observed, where local values differ markedly from their surrounding neighborhoods. The spatial distribution of these clusters and outliers varies considerably across the study area, demonstrating that spatial dependence is not constant over space. This confirms that the underlying spatial process governing observation density is non-stationary.

Table I summarizes the spatial distribution of Local Moran's I cluster types, showing that significant clustering occurs only in localized regions, confirming spatial non-stationarity in observation density.

C. Global Spatial Autocorrelation

To establish a baseline understanding of spatial dependence before exploring local variations, global spatial autocorrelation measures were computed. The Global Moran's I statistic yielded a value of 0.8118 ($p \leq 0.001$), indicating a highly significant positive spatial autocorrelation across the dataset. This confirms that cells with similar ecological values are strongly clustered globally. Furthermore, the Global Geary's C statistic was calculated as 0.2319 ($p \leq 0.001$). As Geary's C values less than 1 denote positive spatial association, this further substantiates the presence of a strong global spatial structure in the migratory species data.

D. Getis-Ord G_i^* Hotspot Analysis

To complement the LISA analysis, the Local Getis-Ord G_i^* statistic was employed to identify statistically significant spatial hotspots and coldspots (Fig. 7). The G_i^* cluster map corroborates the localized spatial non-stationarity observed in the LISA map. Distinct Hotspots (regions of significantly high values) are clearly delineated in the northern and western regions, whereas profound Coldspots (regions of significantly low values) dominate the central interior. This dual-method approach confirms that migratory observation patterns are characterized by significant, spatially varying dependence structures.

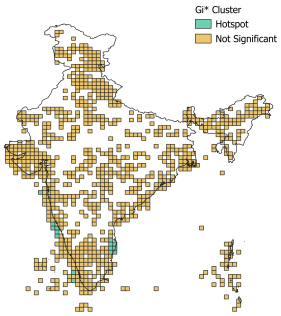

Fig. 7. Getis-Ord G_i^* cluster map identifying statistically significant spatial hotspots (blue) and not significant places (yellow) in the study area.

E. Discussion and Implications

The spatial statistical methods applied in this study successfully capture both global and local spatial characteristics of migratory species observation data. Spatial heterogeneity analysis highlights clear regional variability, while local spatial autocorrelation analysis reveals spatially varying dependence structures. However, due to sparseness of data, many of the outputs were blank, or closer to a value of zero.

The observed non-stationarity suggests that global models assuming constant spatial relationships may be insufficient for fully characterizing migratory species observation patterns. More advanced spatial and spatio-temporal modeling approaches, such as spatial regression or geographically weighted regression, could improve understanding by explicitly accounting for spatial dependence and spatially varying relationships.

IV. SPATIAL BIODIVERSITY MODELLING

While exploratory spatial analysis establishes the presence of spatial dependence and non-stationarity, it does not directly estimate ecological richness in unsampled regions. To address this limitation, predictive spatial models were constructed to estimate expected species richness across the study area. The objective of this stage was not only to explain spatial patterns

but to reconstruct ecological biodiversity distribution beyond observed sampling effort.

A. Spatial Feature Engineering

To incorporate geographic influence into predictive modelling, a neighbourhood feature was constructed using spatial adjacency relationships. A Queen contiguity spatial weights matrix W was defined such that adjacent grid cells sharing either edges or vertices were considered neighbours.

Neighbourhood richness was computed as:

$$N_i = \sum_j W_{ij} y_j \tag{1}$$

where y_j denotes species richness in neighbouring regions. This feature represents ecological context and enables models to learn spatial continuity rather than treating locations as independent samples.

Each grid cell was therefore represented by the following predictors:

- Observation Density (D_i)
- Migration Ratio (M_i)
- Neighbour Richness (N_i)

B. Spatial Negative Binomial Regression

Species richness is a count variable exhibiting overdispersion; therefore a Negative Binomial Generalised Linear Model (GLM) with log link was employed:

$$\log(E[y_i]) = \beta_0 + \beta_1 D_i + \beta_2 M_i + \beta_3 N_i \tag{2}$$

This model captures interpretable ecological relationships while accounting for spatial autocorrelation through the neighbourhood term.

C. Machine Learning Models

To capture non-linear ecological relationships, two ensemble machine learning models were evaluated.

1) Random Forest Regression: Random Forest regression constructs multiple decision trees and averages predictions, enabling modelling of non-linear interactions between ecological variables and neighbouring habitats.

2) Gradient Boosting Regression: Gradient Boosting iteratively improves prediction error by sequentially fitting trees to residuals, allowing the model to capture smooth spatial gradients in biodiversity.

D. Model Evaluation

Models were evaluated using spatial hold-out validation, where a subset of grid cells was removed during training and predicted using surrounding ecological information.

The Random Forest model achieved the highest predictive performance, indicating biodiversity relationships are strongly non-linear. The statistical spatial model, while less accurate, remains valuable for ecological interpretability.

TABLE II
PREDICTIVE MODEL PERFORMANCE COMPARISON

Model	RMSE	MAE	R^2
Spatial Negative Binomial	0.507	0.35	0.13
Random Forest	0.467	0.242	0.33
Gradient Boosting	0.493	0.276	0.17

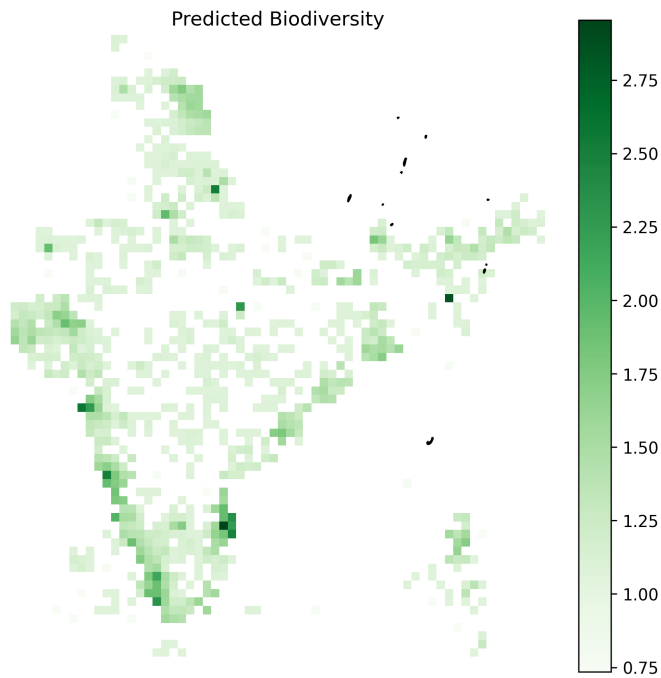

Fig. 8. Predicted biodiversity richness estimated using a spatial machine learning model.

E. Predicted Biodiversity Distribution

The trained models were used to estimate expected biodiversity across all grid cells, producing a continuous ecological richness map (Fig. 8). Unlike raw observation density, the predicted map reflects ecological potential rather than sampling effort.

F. Residual Hotspot Analysis

To identify ecological anomalies, residuals were computed as:

$$Residual_i = Observed_i - Predicted_i \tag{3}$$

Positive residuals indicate regions richer than expected, while negative residuals indicate under-represented ecosystems. The residual map (Fig. 9) highlights potential conservation priority areas.

G. Interpretation

The modelling results demonstrate that biodiversity is primarily governed by the surrounding ecological context rather than isolated local measurements. The strong performance of the Random Forest model indicates biodiversity distribution

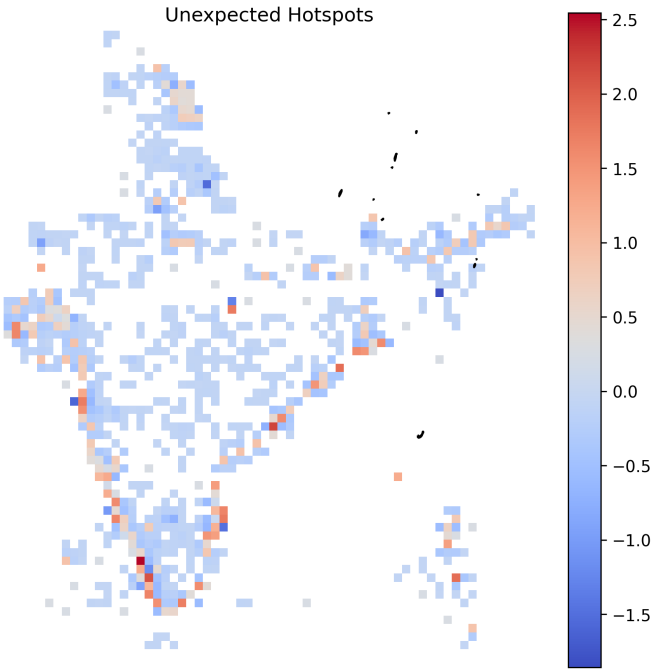

Fig. 9. Residual hotspot map showing regions richer or poorer than predicted ecological expectation.

follows non-linear spatial patterns consistent with ecological habitat transitions.

Importantly, the predicted map differs significantly from raw observation density, confirming that sampling bias strongly affects recorded biodiversity data.

V. Spatial Biodiversity Modelling

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The trained models were used to estimate expected biodiversity across all grid cells, producing a continuous ecological richness map (Fig. 8). Unlike raw observation density, the predicted map reflects ecological potential rather than sampling effort.

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G. Interpretation

The modelling results demonstrate that biodiversity is primarily governed by the surrounding ecological context rather than isolated local measurements. The strong performance of the Random Forest model indicates biodiversity distribution follows non-linear spatial patterns consistent with ecological habitat transitions.

Importantly, the predicted map differs significantly from raw observation density, confirming that sampling bias strongly affects recorded biodiversity data.

Part Contribution

1) **Ahil** → Data pre-processing, Feature Engineering, Model Training (Task 4)
2) **Darsh** → Data pre-processing, ESDA, Spatial Autocorrelation (Task 2)
3) **Vinay** → Spatial Stationarity and Heterogeneity (Task 3), Report Generation

Declaration of GenAI Usage

We used LLMs such as Gemini AI and ChatGPT for analysing the columns and what they represent in the dataset. We used them to generate code for specific parts of the training model, and for the use of the geopandas Python library in the code. The rest of the code and discussion is done by the authors.